```
import pandas as pd

import matplotlib.pyplot as plt

from sklearn.tree import DecisionTreeClassifier, plot_tree

from sklearn.preprocessing import LabelEncoder

# 1. Load dataset

df = pd.read_excel("../../data/Online Retail.xlsx")

# 2. Preprocessing

df = df.dropna(subset=['CustomerID'])

df = df[

    (df['Quantity'] > 0) &

    (df['UnitPrice'] > 0)

]

# 3. Calculate total spend

df['TotalSpend'] = df['Quantity'] * df['UnitPrice']

# 4. Group data by customer

customer_df = df.groupby('CustomerID').agg({

    'Quantity': 'sum',

    'UnitPrice': 'mean',

    'TotalSpend': 'sum',

    'Country': 'first'

}).reset_index()

# 5. Create target variable

threshold = customer_df['TotalSpend'].median()

customer_df['HighValue'] = (

    customer_df['TotalSpend'] > threshold

).astype(int)

# 6. Encode country

label = LabelEncoder()

customer_df['Country'] = label.fit_transform(

    customer_df['Country'] )

# Select features and target

X = customer_df[['Quantity', 'UnitPrice', 'Country']]

y = customer_df['HighValue']

# 7. Train entropy-based decision tree

model = DecisionTreeClassifier(

    criterion='entropy',

    max_depth=4,

    random_state=42

)

model.fit(X, y)

# 8. Visualize decision tree

plt.figure(figsize=(18, 8))

plot_tree(

    model,

    feature_names=X.columns,

    class_names=['Low', 'High'],

    filled=True,

    rounded=True

)

plt.title("Customer Segmentation using Decision Tree")

plt.show()

# 9. Display feature importance

importance = pd.Series(

    model.feature_importances_,

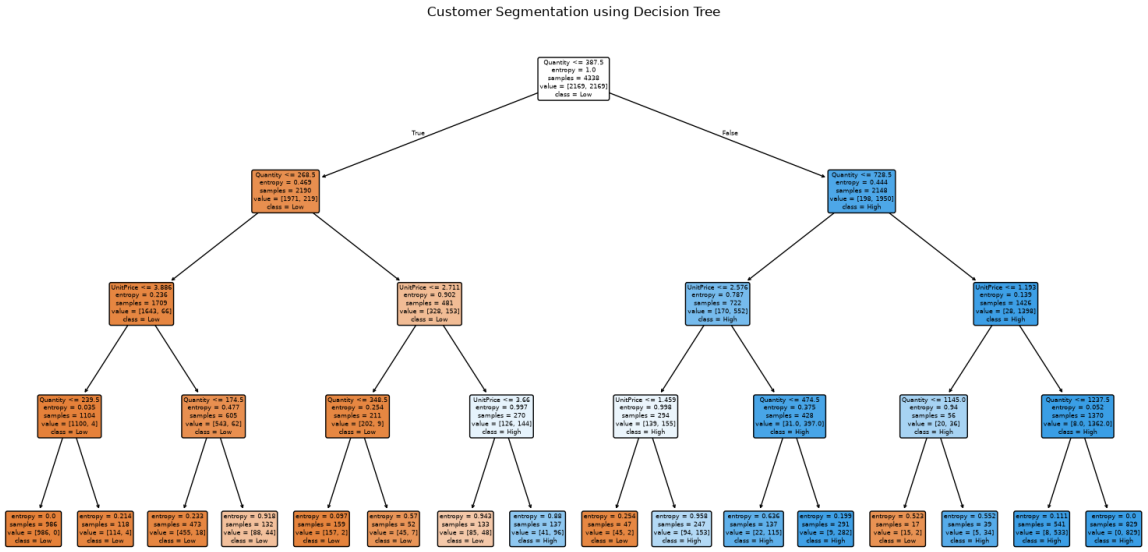
    index=X.columns

)

print("Feature Importance:")

print(importance.sort_values(ascending=False))
```

```
Feature Importance:
Quantity     0.86527
UnitPrice    0.13473
Country      0.00000
dtype: float64
```



Exp10 output

